

## Birth

Each stage of a star's life cycle takes place over billions of years. All stars are born from giant gas and dust clouds, called nebulae. Gravity pulls the gas and dust together to form hot, spinning clumps.



## Young and bright

Over time, these clumps get so hot that nuclear reactions start inside them, and they turn into stars. The intense heat makes them glow. A young star is called a protostar—*proto* means *early*.

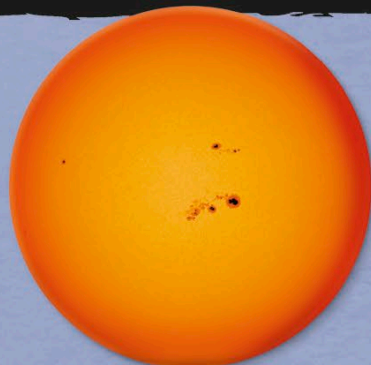
## Death

The planetary nebula drifts away and is scattered into space, where it will in time form new stars. All that is left of the star is the glowing core, called a white dwarf. This gets smaller until it cools and becomes a black dwarf.

# Star

Just like animals, stars are born, grow and develop through life, and then die. Depending on its type and size, a star can live and die in different ways. Bigger stars shine more brightly but live shorter lives than smaller stars. This is the life cycle of a medium-sized star.

**The sun** The sun is a medium-sized star and will follow the same life cycle as the one above. The sun formed about 4.6 billion years ago and is presently at the main-sequence stage. In about 5 billion years, it will die.



**Black dwarf** Stars die in different ways. Medium-sized stars become dark, dead blobs called black dwarfs. Bigger stars collapse into themselves completely. They turn into black holes. Here, gravity is so strong that no light can escape.

